

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

POWER STEERING CONTROL MODULE (PSCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital

multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.

- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.
- BEFORE starting any repairs or investigations of power steering issues, first download and retain the SDD session file using the manufacturer-approved diagnostic system. Where possible, check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms.
- During normal vehicle operation steering assistance may be temporarily reduced in a controlled way under some circumstances. When diagnosing potential steering assistance issues, first check if the customer reported symptom relates **either** to one of these circumstances (as described in the relevant Description and Operation section in the workshop manual, see Section 211), **or**, to one of the following circumstances: an over temperature condition due to exceptionally demanding steering inputs; the vehicle power supply being in a low state of charge; CAN network signals required from other vehicle systems being unavailable or invalid

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Power Steering Control Module (PSCM). For additional

diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Power Steering](#) (211-02 Power Steering, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1304-04	Electronic Power Assisted Steering System - System Internal Failures	NOTE: This DTC will log with the ignition set to ON ▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Steering gear internal failure	NOTE: Once repair is complete, clear DTC's, close session and then upload SDD session file to JLR. Failure to follow this instruction may result in a warranty claim becoming invalid ▪ Using the manufacturer approved diagnostic system, check datalogger signals - Main ECU Supply Voltage (0xDD02) - Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits and associated connectors for open circuit, high resistance. Repair or replace circuit(s)/connector(s) as required. Clear DTC and retest ▪ Check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms. If the time of DTC occurrence does not relate to the customer reported symptom, check for other appropriately timed and relevant DTCs in all vehicle modules. If the outcome of these

			investigations is not conclusive, contact the JLR Technical Helpdesk ▪ If the steering gear fault persists, check and install a new steering gear as required
B1304-07	Electronic Power Assisted Steering System - Mechanical failures	NOTE: This DTC will log with the ignition set to ON ▪ Incorrect wheel or tire sizes ▪ Incorrect tire pressures ▪ Steering or suspension systems mechanical faults ▪ Incorrect steering calibration load to vehicle	▪ Refer to the relevant section of the workshop manual and check the wheel and tire sizes ▪ Refer to the relevant section of the workshop manual and check the tire pressures ▪ Refer to the relevant section of the workshop manual and check steering system for mechanical faults or wear, rectify as required ▪ Refer to the relevant section of the workshop manual and check suspension system for mechanical faults, rectify as required ▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software to the power steering control module. Clear the DTCs and retest
B1304-16	Electronic Power Assisted Steering System - Circuit voltage below threshold	NOTE: This DTC will log with the ignition set to ON ▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	

			NOTE: The system is designed to give a heavier but stable steering feel when a low voltage is detected. It is recommended to reduce drastic inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate, high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert
			▪ Using the manufacturer approved diagnostic system, check datalogger signals - Main ECU Supply Voltage (0xDD02) - Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits and associated connectors for open circuit, high resistance. Repair or replace circuit(s)/connector(s) as required. Clear DTC and retest ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ If the fault persists and no battery or charging issues can be identified, contact the JLR Technical Helpdesk. Do not replace the steering gear
B1304-17	Electronic Power Assisted Steering System - Circuit		▪ Using the manufacturer approved diagnostic system, check datalogger signals -

	voltage above threshold	NOTE: This DTC will log with the ignition set to ON ▪ Battery/charging system fault	Main ECU Supply Voltage (0xDD02) - Vehicle Battery Voltage (0x402A). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ If the fault persists and no battery or charging issues can be identified, contact the JLR Technical Helpdesk. Do not replace the steering gear
B1304-98	Electronic Power Assisted Steering System - Component or system over temperature	NOTE: This DTC will log with the ignition set to ON ▪ Power steering control module over temperature due to hot climate or excessive steering inputs by customer	NOTE: The system is designed to give a heavier but stable steering feel when high temperature is detected to protect the system from permanent damage. It is recommended to reduce level of steering inputs to the steering in the specific scenarios identified by the customer to provoke the problem. Depending on the level of reduction this could be accompanied by an alert ▪ Check for heat shield damage or mud packing around power steering control module. Allow the power steering control module to cool. Consider the environmental temperature ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest
C102D-07	High Friction Inside Power Steering -		

Mechanical
failures

NOTE:

This DTC will
log with the
ignition set to
ON

- Abnormally high friction in steering ball joints or suspension components
- Abnormally high friction in steering gear

NOTES:

- Do not damage bellows on steering gear removal. Failure to follow this instruction may result in warranty becoming invalid for steering gear assembly
- DO NOT reset DTCs and record SDD session then load to the internet. Failure to follow this instruction may result in warranty claim becoming invalid
- If this DTC is logged and BEFORE starting any repairs or further investigations of power steering issues, first download and retain the SDD session file using the manufacturer-approved diagnostic system. Where possible, check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms

- Disconnect the steering gear from the suspension. Check the suspension system for freedom of movement or abnormally high friction
- Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of

			water ingress is present, replace the steering gear. Refer to the new module/component installation note at the top of the DTC index ▪ Check the steering gear for abnormally high friction. If abnormally high friction is found, check if the onset of the steering system fault symptoms occurred at the same time that the DTC was logged. If the customer reports of heavy steering coincide with the logging of this DTC, then install new steering gear as required ▪ If the DTC was logged between medium and high time (medium to high mileage) in service it is most likely caused by impact or water ingress damage. Please refer to - Electric Power Steering Gear Damage Assessment Checklist – in section 211-00/02 Diagnosis and Testing of the workshop manual
C200B-01	Steering Shaft Torque Sensor 1 - General electrical failure	NOTE: This DTC will log with the ignition set to ON ▪ Power steering control module torque sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Power steering control module torque sensor	NOTE: DO NOT reset DTCs and record SDD session then load to the internet. Failure to follow this instruction may result in warranty claim becoming invalid ▪ Refer to the electrical circuit diagrams and check the security and integrity of the power steering control module torque sensor circuit cable and connectors. Rectify as required ▪ Using the manufacturer

		internal failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, check and install a new steering gear as required
C200B-31	Steering Shaft Torque Sensor 1 - No signal	NOTE: This DTC will log with the ignition set to ON - Power steering control module torque sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Power steering control module torque sensor internal failure	NOTE: DO NOT reset DTCs and record SDD session then load to the internet. Failure to follow this instruction may result in warranty claim becoming invalid - Refer to the electrical circuit diagrams and check the power steering control module torque sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check and install a new steering gear as required
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	NOTE: This DTC will log with the ignition set to ON Invalid data received from another control module via the high speed CAN bus (chassis)	Using the manufacturer approved diagnostic system, check the snap shot data to determine the invalid data source control module. Check the relevant control for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication		Using the manufacturer approved diagnostic system,

	Bus - Alive / sequence counter incorrect / not updated	NOTE: This DTC will log with the ignition set to ON Invalid data received from another control module via the high speed CAN bus	check the snap shot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	NOTE: This DTC will log with the ignition set to ON Missing message from another control module via the high speed CAN bus (chassis)	Using the manufacturer approved diagnostic system, check the snap shot data to determine the missing message source control module, check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	NOTE: This DTC will log with the ignition set to ON High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-56	Internal Control Module Software Incompatibility -		Check for related DTCs stored in other modules. Check that the correct body

	Invalid/incomplete configuration	NOTE: This DTC will log with the ignition set to ON - Incorrect body control module/gateway module assembly power or ground circuit open circuit, high resistance installed to the vehicle - Incorrect power steering control module installed to the vehicle	control module/gateway module assembly is installed to the vehicle. Using the manufacturer approved diagnostic system check and install latest relevant level of software to the body control module/gateway module assembly Check that the correct power steering control module is installed to the vehicle. Using the manufacturer approved diagnostic system check and install latest relevant level of software to the power steering control module
U2012-87	Car Configuration Parameter(s) - Missing message	NOTE: This DTC will log with the ignition set to ON - Car configuration file mismatch with vehicle specification - Power steering control module is not configured correctly - Incorrect central junction box installed to the vehicle	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. - Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, reconfigure the power steering control module with the latest level software - Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC

			index
U2100-00	Initial Configuration Not Complete - No sub type information	NOTE: This DTC will log with the ignition set to ON ▪ Car configuration file mismatch with vehicle specification ▪ Power steering control module is not configured correctly ▪ Incorrect central junction box installed to the vehicle	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, reconfigure the power steering control module with the latest level software ▪ Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index
U2100-56	Initial Configuration Not Complete - Invalid/incomplete configuration	NOTE: This DTC will log with the ignition set to ON ▪ Power steering control module is not configured correctly	▪ Using the manufacturer approved diagnostic system, reconfigure the power steering control module with the latest level software. Using the manufacturer approved diagnostic system, perform routine - Lock Steering Tune To Vehicle (0x3038)
U2101-00	Control Module Configuration Incompatible - No		

	sub type information	NOTE: This DTC will log with the ignition set to ON ▪ Car configuration file mismatch with vehicle specification ▪ Power steering control module is not configured correctly ▪ Incorrect central junction box installed to the vehicle	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, reconfigure the power steering control module with the latest level software ▪ Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index
U3001-00	Control Module Improper Shutdown - No sub type information	NOTE: This DTC may set when the vehicle battery is disconnected, whilst the ignition is in the 'on' position. This is not a fault. Clear the DTC and retest ▪ Occurrence of an unexpected power steering control module system reset	

NOTES:

- In some case this can be induced during system reduction functions. Ignore if set in conjunction with other DTCs
- If this DTC is logged and BEFORE starting any repairs or further investigations of power steering issues, first download and retain the SDD session file using the manufacturer-approved diagnostic system. Where possible, check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms. Ignore this DTC unless customer complains of heavy or intermittent power steering assistance

- Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance. Clear the DTCs and retest

